Laplace–Resolvent Semi-Analytics for a Ratio Barrier Payoff under Kou Dynamics

April 27, 2026

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1 Introduction

Decentralized finance (DeFi) lending protocols such as AAVE and Compound have opened a new class of capital-efficient trading strategies: a user deposits a cryptocurrency as collateral, borrows a second asset against it, and simultaneously sells the borrowed asset short. The result is a leveraged long–short position whose performance depends on the relative price dynamics of the two assets. With combined total value locked across major DeFi platforms exceeding $10 billion, these strategies have become a meaningful component of on-chain market activity, yet their risk profile remains analytically under-studied.

The defining feature of DeFi lending is automatic liquidation. Each platform computes a real-time *health factor* — a ratio of collateral value to outstanding liability, scaled by the borrowing power of the collateral — and liquidates the position the instant this ratio falls below one. Unlike traditional margin accounts, where a broker may issue a call and allow time for top-up, smart-contract liquidation is instantaneous and irrevocable. A practitioner must therefore choose the initial coin holdings carefully: a more aggressive position amplifies expected returns on surviving paths but raises the probability of forced exit. The central question this paper addresses is how to set those holdings optimally given the underlying asset dynamics.

Wrong-way risk is the dominant concern. An adverse move in either leg — a fall in the collateral asset or a rise in the borrowed asset — simultaneously erodes the numerator and inflates the denominator of the health factor. Cryptocurrency markets are particularly exposed: daily log-returns of 10–20% are common

and flash-crash episodes can exceed 50%, so fat-tailed jump dynamics are essential for capturing realistic liquidation probabilities.

We model the bivariate log-price process under the *Kou double-exponential jump–diffusion* framework (Kou, 2002), in which each asset follows a diffusion with correlated Brownian parts and independent compound Poisson jumps with double-exponentially distributed sizes. This specification captures the asymmetric, leptokurtic return distributions observed in cryptocurrency markets while remaining analytically tractable. In particular, exponential jump sizes are eigenfunctions of the generator, which allows the Laplace-domain resolvent equation to be solved in closed form.

Related literature. This paper is closest to the literature on first-passage and barrier problems under double-exponential jump diffusion, and to the emerging empirical literature on DeFi liquidation. The economic motivation is the mechanical liquidation rule embedded in overcollateralized lending protocols such as AAVE: once the collateral-to-liability ratio falls below the protocol threshold, the position is forcibly and irrevocably unwound. Qin, Zhou, Gamito, Jovanovic, and Gervais (2021) document the incentives, risks, and instability channels associated with such liquidations across major DeFi lending platforms, while Heimbach and Huang (2024) study leverage formation and amplification in DeFi lending markets. These papers establish that liquidation is economically significant but are empirical and institutional in nature. They do not address the ex-ante problem of choosing the initial health factor of a leveraged long–short position under a stochastic liquidation boundary.

The technical core of this paper belongs to the first-passage and barrier-option literature for jump diffusions. Kou and Wang (2003) derive explicit Laplace-transform results for first-passage times of a double-exponential jump diffusion, exploiting the overshoot properties of exponential jumps. Sepp (2004) extends the transform approach to single- and double-barrier option pricing under double-exponential jump diffusion by converting the time-dependent pricing equation into a Laplace-domain boundary-value problem that reduces to a finite algebraic system. Our analysis inherits this structural advantage: exponential jump sizes are eigenfunctions of the Kou generator, turning the integro-differential resolvent equation into a finite-dimensional linear system. The difference is that the payoff studied here is not a standard barrier option on a single asset. It is a barrier-killed long–short ratio payoff generated by a DeFi borrowing position, in which the short-leg factor and the liquidation boundary on the log-ratio are entangled. An exponential change of measure absorbs the short-leg factor and reduces each conditional payoff moment to a one-dimensional first-passage problem for the log-ratio process under a tilted measure. This measure reduction is the paper's main technical step. Time-domain quantities are recovered by numerical Laplace inversion via the Talbot contour algorithm (Talbot, 1979).

This paper is also related, more remotely, to long–short portfolio construction and portfolio optimization under jump risk. Jacobs, Levy, and Starer (1999) emphasise that the long and short legs of a portfolio should be optimised jointly rather than constructed sequentially; here the strategy is static after initiation, so the economically relevant control is the initial log health factor rather than a dynamic rebalancing rule. Portfolio optimization under jump risk is a large literature studying dynamic rebalancing and HJB-type controls, including the mean–variance approach of Dang and Forsyth (2016); our setting differs because the central constraint is an endogenous first-passage liquidation event rather than a soft variance or execution-cost bound.

Measure change. Computing the moments of the payoff conditional on survival requires tracking the joint evolution of both asset prices up to the random liquidation time — a substantially harder object than a one-dimensional counterpart. A change of measure eliminates this complexity. By exponentially tilting the path measure by a power of the short-leg price, that power is absorbed into the new numeraire, so that the expectation under the tilted measure depends only on the log-ratio of the two prices. The bivariate killing problem thus reduces to a *univariate* first-exit problem for the log-ratio process, which remains a Lévy process under the tilted measure. This reduction is purely a matter of analytical tractability: no information about the joint dynamics is discarded, since the characteristic exponent of the tilted log-ratio encodes its full marginal law.

Laplace transform and the resolvent equation. With the problem reduced to one dimension, the killed expectation satisfies a partial integro-differential equation (PIDE) in time and the log-ratio state.

Following Sepp (2004), applying the Laplace transform in time converts this evolution equation into a stationary *ordinary* integro-differential equation (OIDE) in the log-ratio alone, with the Laplace parameter playing the role of a discount rate and the payoff function appearing as the source term. The Kou structure then reduces the OIDE to an effective ODE: because exponential functions are eigenfunctions of the Kou generator, every term in the integro-differential operator acts algebraically on an exponential ansatz, and the resolvent equation becomes a finite-dimensional algebraic system. Boundary conditions at the liquidation barrier supply three equations for the three admissible decaying modes, closing the system. Time-domain quantities are finally recovered by numerical Laplace inversion.

The key object is the *initial log health factor*, the logarithm of the borrowing-power-adjusted ratio of initial collateral value to initial liability. It encodes the initial distance of the position from the liquidation boundary: a larger value means the position starts further from forced exit. Once the net book value is normalised to one unit of account at inception, both coin holdings are determined entirely by this single scalar, reducing the optimisation to a scalar search. The optimal value maximises a penalised conditional mean–variance criterion: the conditional expected payoff on surviving paths, minus a risk-aversion-weighted penalty on its conditional variance, minus a penalty on the probability of liquidation. This balances risk-adjusted return against the cost of forced exit.

The paper makes the following contributions. First, we show that the portfolio optimisation reduces to a one-dimensional scalar problem parameterised by the initial log health factor, with all performance quantities expressible as explicit functions of that single parameter (Section 2). Second, we derive a semi-analytical Laplace–resolvent for killed moments of all admissible orders under the exponentially tilted measure (Section 5): because the payoff kink lies strictly below the liquidation barrier whenever the borrowing power is strictly less than one, the source term is a polynomial on the entire survival domain and the resolvent reduces to a three-by-three linear system with no interface conditions, regardless of moment order. Third, we derive the survival probability by the same resolvent method at zero tilt (Section 5, Subsection 5.2).

The remainder of the paper is organised as follows. Section 2 introduces the position, health factor, payoff, and the one-dimensional reduction to a scalar optimisation. Section 3 specifies the asset dynamics under the Kou model; Section 3.2 introduces the tilted-measure machinery. Section 4 shows how the change of measure decouples the short-leg factor from the barrier problem, reducing each moment to a one-dimensional killed expectation, and states the objective in terms of those moments. Section 5 derives the respective resolvents. Section 6 presents numerical results, comparing the Laplace–resolvent method against Monte Carlo simulation across a range of initial log health factor values and verifying the analytical no-shorting limit.

2 Problem Setup

We consider a leveraged long–short portfolio in which an investor holds a long position in asset S_1 and a short position in asset S_2, financed so that the net initial value of the book equals one unit of account. The strategy is subject to a risk-management rule that forces liquidation if the long leg falls sufficiently relative to the short leg, reflecting margin requirements or stop-loss constraints common in practice. The objective is to choose the initial coin holdings so as to optimise a performance criterion defined on the terminal payoff, conditional on surviving the liquidation threshold.

2.1 Terminal payoff of the long-short portfolio

Let $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, \mathbb{P})$ be a filtered probability space. Two strictly positive asset prices S_1, S_2 evolve as

$$S_{i,t} = S_{i,0} e^{X_{i,t}}, \qquad X_{i,0} = 0, \quad i = 1, 2,$$

where X_1, X_2 are log-price processes to be specified in Section 3. The investor holds $n_1 > 0$ units of asset S_1 (long leg) and $n_2 > 0$ units of asset S_2 (short leg).

Fix $b \in (0, 1)$, the *borrowing power* (or *liquidation threshold* in AAVE's terminology) of the collateral asset S_1: it is the maximum fraction of the collateral value that the protocol requires to be maintained against the outstanding liability at all times. The *health factor* is

$$H_t := \frac{b\, n_1 S_{1,t}}{n_2 S_{2,t}}, \qquad h_t := \log H_t,$$

where h_t is the *log health factor*. Liquidation is triggered the instant the collateral can no longer cover the liability at rate b:

$$\tau := \inf\{t \geq 0 : \; H_t < 1\}.$$

When $H_t \geq 1$ the position is solvent; when $H_t < 1$ the smart contract liquidates it instantaneously and irrevocably. A larger b sets the liquidation boundary more conservatively, triggering early exit on adverse price moves and limiting the protocol's risk of bad debt.

Since

$$h_t = \log\Big(\frac{b\, n_1 S_{1,0}}{n_2 S_{2,0}}\Big) + (X_{1,t} - X_{2,t}),$$

define the initial log health factor and the log-ratio process by

$$h_0 := \log\Big(\frac{b\, n_1 S_{1,0}}{n_2 S_{2,0}}\Big), \qquad Z_t := X_{1,t} - X_{2,t},$$

so that $h_t = h_0 + Z_t$ and the liquidation time is equivalently

$$\tau = \inf\{t \geq 0 : \; Z_t < -h_0\}.$$

The one-dimensional process Z thus inherits all the relevant risk: the position survives if and only if Z stays above the level $-h_0$.

The gross profit at horizon T, contingent on survival, is

$$\Pi_T := (n_1 S_{1,T} - n_2 S_{2,T})^+ \, \mathbf{1}_{\{\tau > T\}}.$$

The positive part reflects that the investor closes the position only if it is in profit at T; the indicator $\mathbf{1}_{\{\tau>T\}}$ reflects that the position must have survived to maturity. Using the definition of H_T, one obtains the following factorization.

Lemma 2.1 (Payoff factorization)**.**

$$\Pi_T = \frac{n_2}{b}\, S_{2,0} e^{X_{2,T}} \, (e^{h_0 + Z_T} - b)^+ \, \mathbf{1}_{\{\tau > T\}}. \tag{1}$$

This separates the payoff into a numeraire factor driven by S_2, and a barrier-killed option on the log-ratio Z with moneyness controlled by h_0.

2.2 Optimization setup

The two coin holdings (n_1, n_2) represent two degrees of freedom. A normalisation reduces this to a single scalar, and the structure of the payoff then identifies h_0 as the natural optimisation variable.

We impose the unit-initial-value constraint

$$n_1 S_{1,0} - n_2 S_{2,0} = 1,$$

which fixes the net book value at inception. Setting $x := n_2 S_{2,0}$, one has $n_1 S_{1,0} = 1 + x$ and $e^{h_0} = b(1+x)/x$. Solving for x,

$$x = \frac{b}{e^{h_0} - b},$$

which yields the holding map

$$n_2(h_0) = \frac{b}{S_{2,0}(e^{h_0} - b)}, \qquad n_1(h_0) = \frac{e^{h_0}}{S_{1,0}(e^{h_0} - b)}. \tag{2}$$

Positivity of both holdings requires $e^{h_0} > b$, i.e.

$$h_0 > \ln b.$$

Optimisation over h_0. Since every quantity of interest — the expected payoff, its variance, and the liquidation probability — is determined by h_0 through (2) and the killed process (Z, τ), the portfolio problem reduces to a one-dimensional optimisation:

$$\max_{h_0 \in (\ln b,\, \infty)} \Phi(h_0)\,,$$

where Φ is a scalar performance criterion, for example the Sharpe ratio, expected return subject to a liquidation probability constraint, or a mean–variance objective. The optimal holdings are then recovered by substituting $h_0^\star$ back into (2).

The specific criterion studied in this paper is a penalised conditional mean–variance objective. Let $\mathbb{E}[\,\cdot \mid \tau > T]$ denote expectation conditional on non-liquidation. The objective is

$$\Phi(h_0) := \mathbb{E}[\Pi_T \mid \tau > T] - \tfrac{\alpha}{2}\,\mathrm{Var}(\Pi_T \mid \tau > T) - \delta\, p_{\mathrm{liq}}(T, 0; h_0),$$

where $\alpha \geq 0$ is a risk-aversion coefficient and $\delta \geq 0$ penalises the probability of forced liquidation. The first two terms reward a high risk-adjusted return on the surviving paths; the third term discourages positions so aggressive that liquidation becomes likely. The interplay between the three terms means that the optimal $h_0^\star$ balances leverage (higher h_0 increases the conditional payoff but also raises liquidation risk) against downside protection.

The remainder of the paper is devoted to computing the building blocks of Φ: the killed moments $m_k(T; h_0)$ and the survival probability $p_{\mathrm{surv}}(T, 0; h_0)$, each as a closed-form function of h_0, evaluated semi-analytically via the Laplace–resolvent method developed in Section 5.

3 Asset Dynamics

3.1 Bivariate Kou model

We assume under $\mathbb{P}$ that each log-price satisfies the SDE

$$\mathrm{d}X_{i,t} = \mu_i^{\mathrm{eff}}\,\mathrm{d}t + \sigma_i\,\mathrm{d}B_{i,t} + \mathrm{d}\left(\sum_{n=1}^{N_{i,t}} J_i^{(n)}\right), \qquad i = 1, 2, \tag{3}$$

where B_1, B_2 are correlated standard Brownian motions with $\mathrm{d}\langle B_1, B_2\rangle_t = \rho\,\mathrm{d}t$, N_i is a Poisson process with intensity λ_i independent of (B_1, B_2), and the jump sizes $J_i^{(n)}$ are i.i.d. with the Kou double-exponential law. The jump density, characteristic function, and moment generating function are

$$f_i(j) = \frac{p_i}{\eta_{i,+}} e^{-j/\eta_{i,+}} \mathbf{1}_{\{j>0\}} + \frac{1-p_i}{\eta_{i,-}} e^{j/\eta_{i,-}} \mathbf{1}_{\{j<0\}}, \qquad \varphi_{J_i}(u) = \frac{p_i}{1 - iu\eta_{i,+}} + \frac{1-p_i}{1 + iu\eta_{i,-}}, \tag{4}$$

$$M_i(s) = \frac{p_i}{1 - s\eta_{i,+}} + \frac{1-p_i}{1 + s\eta_{i,-}}, \qquad s \in \left(-\tfrac{1}{\eta_{i,-}},\, \tfrac{1}{\eta_{i,+}}\right). \tag{5}$$

Here $\eta_{i,+}$ and $\eta_{i,-}$ are the *means* of the positive and negative jump sizes (not exponential rates); the corresponding rates are $1/\eta_{i,+}$ and $1/\eta_{i,-}$. The constraint $\eta_{i,+} < 1$ is required for the price-jump compensator χ_i defined below to be finite.

Drift convention and effective drift. The parameter μ_i is the *continuously compounded expected asset-price growth rate*, i.e. $\mathbb{E}_{\mathbb{P}}[S_{i,t}/S_{i,0}] = e^{\mu_i t}$. Define the price-jump compensator

$$\chi_i := M_i(1) - 1 = \mathbb{E}[e^{J_i} - 1] = \frac{p_i}{1 - \eta_{i,+}} + \frac{1-p_i}{1 + \eta_{i,-}} - 1. \tag{6}$$

Since $\mathbb{E}_{\mathbb{P}}[e^{X_{i,t}}] = \exp(t[\mu_i^{\mathrm{eff}} + \frac{1}{2}\sigma_i^2 + \lambda_i\chi_i])$, matching to $e^{\mu_i t}$ gives the log-price drift coefficient

$$\mu_i^{\mathrm{eff}} := \mu_i - \tfrac{1}{2}\sigma_i^2 - \lambda_i\chi_i. \tag{7}$$

All analytical and numerical objects in the sequel use μ_i^{eff}.

The bivariate process (X_1, X_2) is a Lévy process with joint characteristic exponent

$$\begin{aligned}\Psi(u,v) := i(\mu_1^{\text{eff}} u + \mu_2^{\text{eff}} v) - \tfrac{1}{2}\Big(\sigma_1^2 u^2 + \sigma_2^2 v^2 + 2\rho\sigma_1\sigma_2 uv\Big) \\ + \lambda_1\big(\varphi_{J_1}(u) - 1\big) + \lambda_2\big(\varphi_{J_2}(v) - 1\big),\end{aligned} \tag{8}$$

so that $\mathbb{E}_{\mathbb{P}}[e^{i(uX_{1,t}+vX_{2,t})}] = e^{t\Psi(u,v)}$.

Parameter summary. Table 1 collects all model parameters and their roles.

Symbol	Asset	Economic meaning / role	Constraint
μ_i	$i = 1, 2$	Continuously compounded expected price growth rate: $\mathbb{E}[S_{i,t}/S_{i,0}] = e^{\mu_i t}$	
σ_i	$i = 1, 2$	Diffusion volatility of log-price X_i	$\sigma_i > 0$
ρ	—	Correlation between the two Brownian drivers	$\lvert\rho\rvert < 1$
λ_i	$i = 1, 2$	Jump arrival intensity (expected jumps per unit time)	$\lambda_i > 0$
p_i	$i = 1, 2$	Probability that a jump in X_i is upward	$p_i \in (0, 1)$
$\eta_{i,+}$	$i = 1, 2$	Mean size of upward jumps in X_i	$0 < \eta_{i,+} < 1$
$\eta_{i,-}$	$i = 1, 2$	Mean size of downward jumps in X_i	$\eta_{i,-} > 0$
χ_i	$i = 1, 2$	Price-jump compensator $M_i(1) - 1 = \mathbb{E}[e^{J_i} - 1]$; enters μ_i^{eff} via (7)	
μ_i^{eff}	$i = 1, 2$	Actual drift coefficient in the log-price SDE (3)	
$\Psi(u, v)$	—	Joint characteristic exponent; $e^{T\Psi(0,-ik)} = \mathbb{E}_{\mathbb{P}}[e^{kX_{2,T}}]$ appears in the moment formula	
$S_{i,0}$	$i = 1, 2$	Initial price of asset i	$S_{i,0} > 0$
b	—	Borrowing power of the collateral (AAVE liquidation threshold)	$b \in (0, 1)$
T	—	Investment horizon	$T > 0$
n_i	$i = 1, 2$	Number of coins held: $n_1 > 0$ long in S_1, $n_2 > 0$ short in S_2	$n_i > 0$
H_t	—	Health factor: $b\, n_1 S_{1,t}/(n_2 S_{2,t})$; position liquidated the first time $H_t < 1$	
h_t	—	Log health factor: $h_t = \log H_t = h_0 + Z_t$	
h_0	—	Initial log health factor; the optimisation variable; determines n_i via (2)	$h_0 > \log b$
τ	—	Liquidation time: $\tau = \inf\{t \geq 0 : H_t < 1\}$	
Π_T	—	Terminal payoff: $(n_1 S_{1,T} - n_2 S_{2,T})^+ \mathbf{1}_{\{\tau > T\}}$	

Table 1: Model parameters and derived quantities. The first block contains freely chosen Kou model inputs; the second block contains derived model quantities; the third block contains portfolio and contract inputs; the fourth block contains derived portfolio quantities.

3.2 Tilted Dynamics

For each integer $k \geq 1$, define the k-tilted measure $\mathbb{P}^{(k)}$ by

$$\left.\frac{\mathrm{d}\mathbb{P}^{(k)}}{\mathrm{d}\mathbb{P}}\right|_{\mathcal{F}_t} = \frac{e^{kX_{2,t}}}{\mathbb{E}_{\mathbb{P}}[e^{kX_{2,t}}]} = \exp(kX_{2,t} - t\,\Psi(0, -ik)), \qquad k\eta_{2,+} < 1.$$

Lemma 3.1 (Lévy exponent under $\mathbb{P}^{(k)}$)**.** Under $\mathbb{P}^{(k)}$, the log-ratio $Z = X_1 - X_2$ is again a Lévy process with exponent

$$\psi_Z^{(k)}(s) = \mu_Z^{(k)} s + \tfrac{1}{2}\sigma_Z^2 s^2 + \lambda_1\big(M_1(s) - 1\big) + \lambda_2\big(M_2(k - s) - M_2(k)\big), \tag{9}$$

where

$$\sigma_Z^2 := \sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2 \geq 0, \qquad \mu_Z^{(k)} := (\mu_1^{\text{eff}} - \mu_2^{\text{eff}}) - k(\sigma_2^2 - \rho\sigma_1\sigma_2).$$

In particular, the generator satisfies $\mathcal{L}^{(k)} e^{sz} = \psi_Z^{(k)}(s) e^{sz}$.

This eigenrelation is the key identity used in Section 5 to build the resolvent equation.

Remark 3.1 (Superscript (k) convention)**.** A superscript (k) indicates evaluation under $\mathbb{P}^{(k)}$. The untilted case $k = 0$ recovers the original dynamics of Z under $\mathbb{P}$. For $k \geq 1$, the k-th moment $\mathbb{E}[\Pi_T^k]$ is computed under $\mathbb{P}^{(k)}$; the constraint $k\eta_{2,+} < 1$ ensures the tilted measure and the characteristic root structure remain well-defined.

For the Kou structure, the tilt transforms the jump rates of Z into the k-dependent phase rates

$$r_{1,+}^{(k)} := \frac{1}{\eta_{1,+}}, \quad r_{1,-}^{(k)} := \frac{1}{\eta_{1,-}}, \quad r_{2,+}^{(k)} := \frac{1}{\eta_{2,-}} + k, \quad r_{2,-}^{(k)} := \frac{1}{\eta_{2,+}} - k.$$

Because $Z = X_1 - X_2$, positive jumps of X_2 appear as downward jumps of Z: the tilt by e^{kX_2} changes the downward phase rate from $1/\eta_{2,+}$ to $1/\eta_{2,+} - k$, which is positive precisely when $k\eta_{2,+} < 1$. The associated phase operators integrate f against the exponential jump kernel:

$$K_{j,+}^{(k)}[f](z) := \int_0^\infty f(z+y)\, r_{j,+}^{(k)} e^{-r_{j,+}^{(k)} y}\, \mathrm{d}y, \qquad K_{j,-}^{(k)}[f](z) := \int_{-\infty}^0 f(z+y)\, r_{j,-}^{(k)} e^{r_{j,-}^{(k)} y}\, \mathrm{d}y,$$

for $j = 1, 2$.

Lemma 3.2 (Phase operators on exponentials)**.** For any γ with $\gamma < r_{j,+}^{(k)}$ and $\gamma > -r_{j,-}^{(k)}$,

$$K_{j,+}^{(k)}[e^{\gamma\cdot}](z) = \frac{r_{j,+}^{(k)}}{r_{j,+}^{(k)} - \gamma} e^{\gamma z}, \qquad K_{j,-}^{(k)}[e^{\gamma\cdot}](z) = \frac{r_{j,-}^{(k)}}{r_{j,-}^{(k)} + \gamma} e^{\gamma z}. \tag{10}$$

The nonlocal jump integral thus reduces to a scalar multiple of the same exponential. This is the key algebraic identity exploited in Section 5 to convert the PIDE into a finite-dimensional linear system.

Tilted parameter summary. Table 2 collects the quantities introduced in this subsection.

Symbol	Definition	Role
$\mathbb{P}^{(k)}$	$\mathrm{d}\mathbb{P}^{(k)}/\mathrm{d}\mathbb{P} \propto e^{kX_{2,t}}$	Measure under which moment reduction is carried out
k	Non-negative integer, $k\eta_{2,+} < 1$	Tilting order; $k \geq 1$ for moments, $k = 0$ for survival probability
$\psi_Z^{(k)}(s)$	Lévy exponent of Z under $\mathbb{P}^{(k)}$	Characteristic root equation $\psi_Z^{(k)}(\gamma) = q$ governs the resolvent
σ_Z^2	$\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2$	Variance of the Brownian part of Z; appears in $\psi_Z^{(k)}$
$\mu_Z^{(k)}$	$(\mu_1^{\text{eff}} - \mu_2^{\text{eff}}) - k(\sigma_2^2 - \rho\sigma_1\sigma_2)$	Drift of Z under $\mathbb{P}^{(k)}$; appears in $\psi_Z^{(k)}$
$\mathcal{L}^{(k)}$	Generator of Z under $\mathbb{P}^{(k)}$	$\mathcal{L}^{(k)} e^{sz} = \psi_Z^{(k)}(s) e^{sz}$; used in the resolvent PIDE
$r_{j,\pm}^{(k)}$	Phase rates (see above)	Exponential rates of the jump components of Z under $\mathbb{P}^{(k)}$
$K_{j,\pm}^{(k)}$	Phase operators (see above)	Convert jump integrals to scalars on exponential ansatz

Table 2: Tilted-dynamics quantities. Subscript $j = 1$ refers to the contribution from X_1 jumps; $j = 2$ to that from X_2 jumps (sign-reversed because $Z = X_1 - X_2$).

4 Moments via Measure Change

Computing $\mathbb{E}_{\mathbb{P}}[\Pi_T^k]$ directly under $\mathbb{P}$ is difficult because the payoff couples the joint process (X_1, X_2) with the barrier event $\{\tau > T\}$, producing an expectation that is neither separable nor analytically tractable. The key step is a change of measure that absorbs the $e^{kX_{2,T}}$ factor — which carries the short-leg contribution — into the density, leaving a purely one-dimensional killed expectation for the log-ratio Z under the k-tilted measure $\mathbb{P}^{(k)}$. The moment formula then factorises into the known scalar $e^{T\Psi(0,-ik)} = \mathbb{E}_{\mathbb{P}}[e^{kX_{2,T}}]$ and the barrier-killed expectation $m_k(T; h_0)$ under $\mathbb{P}^{(k)}$, which is tractable via the Laplace–resolvent of Section 5.

Define

$$g_k(z; h_0) := (e^{h_0+z} - b)^k.$$

Since the survival event $\{\tau > T\}$ implies $Z_T > -h_0$, and $b \in (0, 1)$ gives $z_*(h_0) := \ln b - h_0 < -h_0$, we have $e^{h_0+Z_T} > 1 > b$ on $\{\tau > T\}$. The positive part in (1) is therefore redundant on the survival set, and g_k is a polynomial in e^{Z_T}. From (1),

$$\Pi_T^k = \left(\frac{n_2}{b}\right)^k S_{2,0}^k e^{kX_{2,T}}\, g_k(Z_T; h_0)\, \mathbf{1}_{\{\tau>T\}}.$$

$$\mathbb{E}_{\mathbb{P}}[\Pi_T^k] = \left(\frac{n_2}{b}\right)^k S_{2,0}^k\, e^{T\Psi(0,-ik)}\, \mathbb{E}_{\mathbb{P}^{(k)}}\Big[g_k(Z_T; h_0)\, \mathbf{1}_{\{\tau>T\}}\Big]. \tag{11}$$

Proposition 4.1 (Moments of any order)**.** Let $k \geq 1$ be an integer satisfying $k\eta_{2,+} < 1$. Define

$$m_k(T; h_0) := \mathbb{E}_{\mathbb{P}^{(k)}}\Big[g_k(Z_T; h_0)\, \mathbf{1}_{\{\tau>T\}}\Big].$$

Then

$$M_k(h_0) := \mathbb{E}_{\mathbb{P}}[\Pi_T^k] = \frac{e^{T\Psi(0,-ik)}}{(e^{h_0} - b)^k}\, m_k(T; h_0),$$

where $m_k(T; h_0) = U^{(k)}(T, 0; h_0)$ is computed via the Laplace–resolvent method of Section 5. The maximum admissible order is $k_{\max} = \lfloor 1/\eta_{2,+} \rfloor - 1$.

Conditional moments and portfolio objective. Let $p_{\text{surv}}(h_0) := \mathbb{P}(\tau > T)$ and $p_{\text{liq}}(h_0) := 1 - p_{\text{surv}}(h_0)$. Since Π_T already contains the survival indicator $\mathbf{1}_{\{\tau>T\}}$, the conditional k-th moment on surviving paths is

$$\mathbb{E}[\Pi_T^k \mid \tau > T] = \frac{M_k(h_0)}{p_{\text{surv}}(h_0)}, \qquad k \geq 1, \quad k\eta_{2,+} < 1.$$

In particular,

$$\mathbb{E}[\Pi_T \mid \tau > T] = \frac{M_1(h_0)}{p_{\text{surv}}(h_0)}, \qquad \text{Var}(\Pi_T \mid \tau > T) = \frac{M_2(h_0)}{p_{\text{surv}}(h_0)} - \left(\frac{M_1(h_0)}{p_{\text{surv}}(h_0)}\right)^2.$$

The mean–variance–liquidation objective is therefore

$$\Phi(h_0) = \frac{M_1(h_0)}{p_{\text{surv}}(h_0)} - \frac{\alpha}{2}\left[\frac{M_2(h_0)}{p_{\text{surv}}(h_0)} - \left(\frac{M_1(h_0)}{p_{\text{surv}}(h_0)}\right)^2\right] - \delta\, p_{\text{liq}}(h_0), \tag{12}$$

and the portfolio optimization reduces to the one-dimensional problem

$$\max_{h_0 \in (\ln b, \infty)} \Phi(h_0),$$

after which the optimal coin holdings are recovered from (2).

Remark 4.1 (No-shorting limit)**.** As $h_0 \to \infty$ the holding map gives $n_2 S_{2,0} \to 0$ and $n_1 S_{1,0} \to 1$, so liquidation becomes impossible and the portfolio converges to a unit long position in asset 1. For every $k \geq 1$,

$$\lim_{h_0 \to \infty} \mathbb{E}[\Pi_T^k] = \mathbb{E}[e^{kX_{1,T}}] = e^{T\Psi(-ik,0)},$$

which provides a useful diagnostic: the formula for $M_k(h_0)$ must satisfy this limit exactly. The $k = 1$ case gives $e^{\mu_1 T}$ by the drift normalisation (7).

The remainder of the analysis is devoted to computing $m_k(T; h_0)$ for all admissible $k \geq 1$ and $p_{\text{surv}}(h_0)$ via the Laplace–resolvent construction.

5 Resolvent Equations for Moments and Survival Probability

This section derives closed-form Laplace-domain resolvents for the killed moments $m_k(T; h_0)$ $(k \geq 1)$ and the survival probability $p_{\text{surv}}(T, 0; h_0)$ $(k = 0)$. Both cases share the same architecture: an exponential ansatz reduces the integro-differential resolvent to a 3×3 linear system whose right-hand side is evaluated at the absorbing barrier.

5.1 Killed Moment Resolvent

Fix an integer $k \geq 1$ with $k\eta_{2,+} < 1$. Define the killed moment function

$$U(t, z) := \mathbb{E}_z^{(k)}\left[g_k(Z_t; h_0)\, \mathbf{1}_{\{\tau(h_0)>t\}}\right], \qquad z > -h_0,$$

with $U(t, z) = 0$ for $z \leq -h_0$. This satisfies the PIDE

$$\partial_t U = \mathcal{L}^{(k)} U, \qquad z > -h_0,$$

with initial condition $U(0, z) = g_k(z; h_0)$. Define the Laplace transform

$$\widehat{U}(q, z) := \int_0^\infty e^{-qt} U(t, z)\, \mathrm{d}t, \qquad q > 0.$$

Applying it to both sides of the PIDE proceeds in two steps.

Left-hand side. Integrate by parts in t:

$$\int_0^\infty e^{-qt}\, \partial_t U(t, z)\, \mathrm{d}t = \left[e^{-qt} U(t, z)\right]_0^\infty + q \int_0^\infty e^{-qt} U(t, z)\, \mathrm{d}t = -U(0, z) + q\widehat{U}(q, z),$$

where the boundary term at $t = \infty$ vanishes by integrability.

Right-hand side. Since $\mathcal{L}^{(k)}$ acts only in z, it commutes with the t-integral:

$$\int_0^\infty e^{-qt}\, \mathcal{L}^{(k)} U(t, z)\, \mathrm{d}t = \mathcal{L}^{(k)} \widehat{U}(q, z).$$

Equating and substituting the initial condition $U(0, z) = g_k(z; h_0)$ gives the resolvent equation

$$(q - \mathcal{L}^{(k)})\widehat{U}(q, z) = g_k(z; h_0), \qquad z > -h_0, \tag{13}$$

with $\widehat{U}(q, z) = 0$ for $z \leq -h_0$.

Assumption 5.1 (Generic six-root regime)**.** Fix an integer $k \geq 0$ satisfying $k\eta_{2,+} < 1$, and let $q > 0$. Assume:

(i) $\sigma_Z > 0$;

(ii) the characteristic equation $\psi_Z^{(k)}(\gamma) = q$ has six simple roots $\gamma_1(q), \ldots, \gamma_6(q)$, ordered so that

$$\mathrm{Re}(\gamma_{1,2,3}) > 0, \qquad \mathrm{Re}(\gamma_{4,5,6}) < 0;$$

(iii) $q \neq \psi_Z^{(k)}(j)$ for $j = 0, \ldots, k$.

Forcing expansion. Since $b \in (0, 1)$ and $z > -h_0$ together imply $e^{h_0+z} > 1 > b$, the payoff kernel is a polynomial in e^z:

$$g_k(z; h_0) = \sum_{j=0}^{k} c_{k,j}(h_0)\, e^{jz}, \qquad c_{k,j}(h_0) := \binom{k}{j} (-1)^{k-j} b^{k-j} e^{jh_0}.$$

Lemma 5.1 (Particular solution)**.** Under Assumption 5.1(iii),

$$\widehat{U}_{\text{part}}(q, z; h_0) = \sum_{j=0}^{k} \frac{c_{k,j}(h_0)}{q - \psi_Z^{(k)}(j)} \, e^{jz} \tag{14}$$

is a particular solution to (13).

Proof. Substitute the forcing expansion into the right-hand side of (13):

$$(q - \mathcal{L}^{(k)})\widehat{U}_{\text{part}} = \sum_{j=0}^{k} c_{k,j}(h_0) \, e^{jz}.$$

Since the sum is finite and $\mathcal{L}^{(k)}$ is linear, it suffices to find, for each j, a function v_j satisfying

$$(q - \mathcal{L}^{(k)})v_j(z) = c_{k,j}(h_0) \, e^{jz}.$$

Propose the exponential ansatz $v_j(z) = A_j \, e^{jz}$. By the eigenrelation (**??**),

$$\mathcal{L}^{(k)}[A_j e^{jz}] = A_j \, \psi_Z^{(k)}(j) \, e^{jz},$$

so

$$(q - \mathcal{L}^{(k)})[A_j e^{jz}] = A_j \big(q - \psi_Z^{(k)}(j)\big) e^{jz}.$$

Matching to $c_{k,j}(h_0) \, e^{jz}$ and using Assumption 5.1(iii) to ensure $q - \psi_Z^{(k)}(j) \neq 0$ gives

$$A_j = \frac{c_{k,j}(h_0)}{q - \psi_Z^{(k)}(j)}.$$

Summing over $j = 0, \ldots, k$ yields (14). □

Lemma 5.2 (Homogeneous solution)**.** Under Assumption 5.1(i)–(ii), the general solution to $(q - \mathcal{L}^{(k)})u = 0$ on $z > -h_0$ that is admissible at $+\infty$ is

$$\widehat{U}_{\text{hom}}(q, z; h_0) = \sum_{m=1}^{3} B_m^{(k)}(q; h_0) \, e^{\gamma_{m+3}(q)(z+h_0)}, \tag{15}$$

where $B_1^{(k)}, B_2^{(k)}, B_3^{(k)}$ are free constants.

Proof. Step 1: exponential ansatz. Seek solutions of the form $u(z) = e^{\gamma z}$. By the eigenrelation (**??**),

$$\mathcal{L}^{(k)} e^{\gamma z} = \psi_Z^{(k)}(\gamma) \, e^{\gamma z},$$

so $(q - \mathcal{L}^{(k)})e^{\gamma z} = (q - \psi_Z^{(k)}(\gamma))e^{\gamma z}$. This vanishes if and only if γ satisfies the characteristic equation

$$\psi_Z^{(k)}(\gamma) = q.$$

Step 2: root count. The exponent $\psi_Z^{(k)}$ is the sum of a quadratic term $\frac{1}{2}\sigma_Z^2\gamma^2$ (non-degenerate by Assumption 5.1(i)) and rational terms from the Kou jumps. Clearing denominators yields a degree-six polynomial in γ, so there are exactly six roots $\gamma_1, \ldots, \gamma_6$ (counted with multiplicity). Under Assumption 5.1(ii) all six are simple and split as $\mathrm{Re}(\gamma_{1,2,3}) > 0$ and $\mathrm{Re}(\gamma_{4,5,6}) < 0$.

Step 3: admissibility. On the half-line $z > -h_0$, the solution must remain bounded as $z \to +\infty$. Since $e^{\gamma z} \to \infty$ whenever $\mathrm{Re}(\gamma) > 0$, the roots $\gamma_1, \gamma_2, \gamma_3$ are inadmissible. Only $\gamma_4, \gamma_5, \gamma_6$ are retained.

Step 4: centering and general solution. Writing $e^{\gamma_{m+3}z} = e^{\gamma_{m+3}(z+h_0)} \cdot e^{-\gamma_{m+3}h_0}$ and absorbing the constant factor into the free coefficient, the general admissible homogeneous solution is

$$\widehat{U}_{\text{hom}}(q, z; h_0) = \sum_{m=1}^{3} B_m^{(k)}(q; h_0) \, e^{\gamma_{m+3}(q)(z+h_0)},$$

where $B_1^{(k)}, B_2^{(k)}, B_3^{(k)}$ are free constants to be determined by the boundary conditions. □

The general solution on $z > -h_0$ is then

$$\widehat{U}(q,z;h_0) = \widehat{U}_{\text{part}}(q,z;h_0) + \widehat{U}_{\text{hom}}(q,z;h_0).$$

Proposition 5.1 (Barrier resolvent)**.** Under Assumption 5.1 and the nondegeneracy condition

$$r_{1,-}^{(k)} \neq r_{2,-}^{(k)}, \tag{16}$$

the boundary conditions at $z = -h_0^+$ uniquely determine $\mathbf{B}^{(k)}(q;h_0) := (B_1^{(k)}, B_2^{(k)}, B_3^{(k)})^\top$ via the linear system

$$\mathbf{M}_{\text{bar}}^{(k)}(q)\,\mathbf{B}^{(k)}(q;h_0) = -\mathbf{b}^{(k)}(q;h_0),$$

where the 3×3 barrier matrix is

$$\mathbf{M}_{\text{bar}}^{(k)}(q) = \begin{pmatrix} 1 & 1 & 1 \\ \dfrac{r_{1,-}^{(k)}}{r_{1,-}^{(k)}+\gamma_4(q)} & \dfrac{r_{1,-}^{(k)}}{r_{1,-}^{(k)}+\gamma_5(q)} & \dfrac{r_{1,-}^{(k)}}{r_{1,-}^{(k)}+\gamma_6(q)} \\ \dfrac{r_{2,-}^{(k)}}{r_{2,-}^{(k)}+\gamma_4(q)} & \dfrac{r_{2,-}^{(k)}}{r_{2,-}^{(k)}+\gamma_5(q)} & \dfrac{r_{2,-}^{(k)}}{r_{2,-}^{(k)}+\gamma_6(q)} \end{pmatrix}$$

and the forcing vector is

$$\mathbf{b}^{(k)}(q;h_0) = \begin{pmatrix} \widehat{U}_{\text{part}}(q,-h_0;h_0) \\ K_{1,-}^{(k)}\Big[\widehat{U}_{\text{part}}(q,\cdot\,;h_0)\Big](-h_0) \\ K_{2,-}^{(k)}\Big[\widehat{U}_{\text{part}}(q,\cdot\,;h_0)\Big](-h_0) \end{pmatrix} = \begin{pmatrix} \displaystyle\sum_{j=0}^{k} \frac{c_{k,j}(h_0)}{q-\psi_Z^{(k)}(j)}\, e^{-jh_0} \\ \displaystyle\sum_{j=0}^{k} \frac{c_{k,j}(h_0)}{q-\psi_Z^{(k)}(j)}\, \frac{r_{1,-}^{(k)}}{r_{1,-}^{(k)}+j}\, e^{-jh_0} \\ \displaystyle\sum_{j=0}^{k} \frac{c_{k,j}(h_0)}{q-\psi_Z^{(k)}(j)}\, \frac{r_{2,-}^{(k)}}{r_{2,-}^{(k)}+j}\, e^{-jh_0} \end{pmatrix}.$$

By Lemma 5.3 below, $\mathbf{M}_{\text{bar}}^{(k)}(q)$ is invertible, giving

$$\mathbf{B}^{(k)}(q;h_0) = -\big(\mathbf{M}_{\text{bar}}^{(k)}(q)\big)^{-1}\mathbf{b}^{(k)}(q;h_0).$$

Proof. Step 1: boundary conditions from killing. The killing condition $\widehat{U}(q,z;h_0) = 0$ for $z \le -h_0$ imposes, at $z = -h_0^+$,

(i) $\widehat{U}(q,-h_0^+;h_0) = 0$;

(ii) $K_{j,-}^{(k)}\Big[\widehat{U}(q,\cdot\,;h_0)\Big](-h_0^+) = 0, \quad j = 1,2.$

Condition (i) is continuity at the barrier. Conditions (ii) arise because $K_{j,-}^{(k)}[f](z) = \int_{-\infty}^{0} f(z+y)\, r_{j,-}^{(k)} e^{r_{j,-}^{(k)} y}\mathrm{d}y$ integrates f over landing points $z+y<z$; for $z = -h_0^+$ every landing point satisfies $z+y < -h_0$, where $\widehat{U} = 0$, so the integral vanishes.

Step 2: substitute the full solution. Write $\widehat{U} = \widehat{U}_{\text{part}} + \widehat{U}_{\text{hom}}$ and insert into each condition. Since the conditions are linear, each splits as

$$[\text{condition on } \widehat{U}_{\text{hom}}] = -[\text{condition on } \widehat{U}_{\text{part}}].$$

Step 3: condition (i). Evaluating at $z = -h_0$ and using $e^{\gamma_{m+3}(q)\cdot 0} = 1$,

$$\widehat{U}_{\text{hom}}(q,-h_0;h_0) = \sum_{m=1}^{3} B_m^{(k)} = -\,\widehat{U}_{\text{part}}(q,-h_0;h_0),$$

which is the first row of the system.

Step 4: conditions (ii). Apply $K_{j,-}^{(k)}$ to $\widehat{U}_{\text{hom}}$. By the phase-operator formula (10),

$$K_{j,-}^{(k)}\left[e^{\gamma_{m+3}(\cdot+h_0)}\right](-h_0) = \frac{r_{j,-}^{(k)}}{r_{j,-}^{(k)}+\gamma_{m+3}(q)}\, e^{\gamma_{m+3}(q)\cdot 0} = \frac{r_{j,-}^{(k)}}{r_{j,-}^{(k)}+\gamma_{m+3}(q)},$$

giving row $j+1$ of the system:

$$\sum_{m=1}^{3} \frac{r_{j,-}^{(k)}}{r_{j,-}^{(k)}+\gamma_{m+3}(q)}\, B_m^{(k)} = -K_{j,-}^{(k)}\left[\widehat{U}_{\text{part}}(q,\cdot\,;h_0)\right](-h_0).$$

Applying (10) to $\widehat{U}_{\text{part}}(q,z;h_0) = \sum_j \frac{c_{k,j}}{q-\psi_Z^{(k)}(j)} e^{jz}$ yields the explicit entries of $\mathbf{b}^{(k)}$ stated above.

Step 5: matrix form and solution. Collecting the three conditions into a single system gives $\mathbf{M}_{\text{bar}}^{(k)}(q)\,\mathbf{B}^{(k)} = -\mathbf{b}^{(k)}$. By Lemma 5.3 below, $\mathbf{M}_{\text{bar}}^{(k)}(q)$ is invertible, so

$$\mathbf{B}^{(k)}(q;h_0) = -\big(\mathbf{M}_{\text{bar}}^{(k)}(q)\big)^{-1}\mathbf{b}^{(k)}(q;h_0). \qquad \square$$

Lemma 5.3 (Invertibility of $\mathbf{M}_{\text{bar}}^{(k)}(q)$)**.** Under Assumption 5.1 and condition (16), the barrier matrix $\mathbf{M}_{\text{bar}}^{(k)}(q)$ of Proposition 5.1 is invertible.

Proof. Subtracting row 1 from rows 2 and 3 replaces entry (i,m) for $i=2,3$ by $-\gamma_{m+3}(q)/(r_{i-1,-}^{(k)}+\gamma_{m+3}(q))$. Pulling out -1 from each of these two rows (net sign $(-1)^2=1$) and then factoring $\gamma_{m+3}(q)$ from each column gives

$$\det\big(\mathbf{M}_{\text{bar}}^{(k)}\big) = \gamma_4\gamma_5\gamma_6\ \det\left(\frac{1}{x_i+\gamma_{j+3}(q)}\right)_{1\le i,j\le 3},$$

where $x_1 := 0$, $x_2 := r_{1,-}^{(k)}$, $x_3 := r_{2,-}^{(k)}$. The right-hand factor is a Cauchy determinant; the Cauchy formula yields

$$\det\left(\frac{1}{x_i+\gamma_{j+3}}\right) = \frac{\prod_{1\le i<i'\le 3}(x_{i'}-x_i)\cdot\prod_{4\le m<m'\le 6}(\gamma_{m'}-\gamma_m)}{\prod_{i=1}^{3}\prod_{m=4}^{6}(x_i+\gamma_m)}.$$

Since $x_1=0$, the factor $\prod_{m=4}^{6}(x_1+\gamma_m) = \gamma_4\gamma_5\gamma_6$ cancels with the prefactor, yielding a ratio whose every factor is nonzero: $r_{1,-}^{(k)} > 0$ since $\eta_{1,-} > 0$; $r_{2,-}^{(k)} > 0$ since $k\eta_{2,+} < 1$; $(r_{2,-}^{(k)} - r_{1,-}^{(k)}) \neq 0$ by (16); the γ_m are pairwise distinct by Assumption 5.1(ii); and $r_{j,-}^{(k)} + \gamma_m \neq 0$ because $-r_{j,-}^{(k)}$ is a pole of $\psi_Z^{(k)}$, so $\psi_Z^{(k)}(-r_{j,-}^{(k)}) = \infty \neq q$, hence $\gamma_m \neq -r_{j,-}^{(k)}$. $\square$

Laplace-domain moment. Evaluating $\widehat{U}$ at the initial state $Z_0 = 0$ gives

$$\widehat{U}(q,0;h_0) = \sum_{j=0}^{k} \frac{c_{k,j}(h_0)}{q-\psi_Z^{(k)}(j)} + \sum_{m=1}^{3} B_m^{(k)}(q;h_0)\, e^{\gamma_{m+3}(q)h_0}. \tag{17}$$

The killed moment is then recovered by Laplace inversion:

$$m_k(T;h_0) = \mathcal{L}_{q\to T}^{-1}\left[\widehat{U}(q,0;h_0)\right].$$

5.2 Survival Probability

The same resolvent architecture yields the survival probability. Define

$$p_{\text{surv}}(t,z;h_0) := \mathbb{P}_z(\tau(h_0) > t), \qquad p_{\text{liq}}(t,z;h_0) := 1 - p_{\text{surv}}(t,z;h_0).$$

Let $\mathcal{L}^{(0)}$ denote the generator of Z under the untilted measure, with exponent $\psi_Z^{(0)}$, and define

$$\widehat{U}_{\text{surv}}(q, z; h_0) := \int_0^\infty e^{-qt} p_{\text{surv}}(t, z; h_0)\, \mathrm{d}t.$$

The problem is

$$(q - \mathcal{L}^{(0)})\widehat{U}_{\text{surv}}(q, z; h_0) = 1, \qquad z > -h_0,$$

with

$$\widehat{U}_{\text{surv}}(q, z; h_0) = 0, \qquad z \le -h_0.$$

The domain is the half-line $(-h_0, \infty)$, the boundary is absorbing at $z = -h_0$, and there is no interface because the source is constant.

Proposition 5.2 (Survival resolvent and liquidation probability)**.** Under Assumption 5.1 and the nondegeneracy condition

$$r_{1,-}^{(0)} \neq r_{2,-}^{(0)}, \quad \text{i.e.,} \quad \eta_{1,-} \neq \eta_{2,+}, \tag{18}$$

let $\gamma_4^{(0)}(q), \gamma_5^{(0)}(q), \gamma_6^{(0)}(q)$ be the three roots of $\psi_Z^{(0)}(\gamma) = q$ with negative real part. Then

$$\widehat{U}_{\text{surv}}(q, z; h_0) = \frac{1}{q} + C_1(q; h_0)e^{\gamma_4^{(0)}(q)(z+h_0)} + C_2(q; h_0)e^{\gamma_5^{(0)}(q)(z+h_0)} + C_3(q; h_0)e^{\gamma_6^{(0)}(q)(z+h_0)},$$

where the coefficients $\mathbf{C}(q; h_0) := (C_1, C_2, C_3)^\top$ satisfy the linear system

$$\mathbf{M}_{\text{surv}}(q)\, \mathbf{C}(q; h_0) = -\begin{pmatrix} 1/q \\ 1/q \\ 1/q \end{pmatrix},$$

with the 3×3 survival matrix

$$\mathbf{M}_{\text{surv}}(q) = \begin{pmatrix} 1 & 1 & 1 \\ \dfrac{r_{1,-}^{(0)}}{r_{1,-}^{(0)} + \gamma_4^{(0)}(q)} & \dfrac{r_{1,-}^{(0)}}{r_{1,-}^{(0)} + \gamma_5^{(0)}(q)} & \dfrac{r_{1,-}^{(0)}}{r_{1,-}^{(0)} + \gamma_6^{(0)}(q)} \\ \dfrac{r_{2,-}^{(0)}}{r_{2,-}^{(0)} + \gamma_4^{(0)}(q)} & \dfrac{r_{2,-}^{(0)}}{r_{2,-}^{(0)} + \gamma_5^{(0)}(q)} & \dfrac{r_{2,-}^{(0)}}{r_{2,-}^{(0)} + \gamma_6^{(0)}(q)} \end{pmatrix}.$$

By Lemma 5.4 below, $\mathbf{M}_{\text{surv}}(q)$ is invertible, giving

$$\mathbf{C}(q; h_0) = -\big(\mathbf{M}_{\text{surv}}(q)\big)^{-1} \begin{pmatrix} 1/q \\ 1/q \\ 1/q \end{pmatrix}.$$

At the initial state $Z_0 = 0$,

$$\widehat{U}_{\text{surv}}(q, 0; h_0) = \frac{1}{q} + C_1(q; h_0)e^{\gamma_4^{(0)}(q)h_0} + C_2(q; h_0)e^{\gamma_5^{(0)}(q)h_0} + C_3(q; h_0)e^{\gamma_6^{(0)}(q)h_0}.$$

Hence

$$p_{\text{surv}}(T, 0; h_0) = \mathcal{L}_{q\to T}^{-1}\Big[\widehat{U}_{\text{surv}}(q, 0; h_0)\Big], \qquad p_{\text{liq}}(T, 0; h_0) = 1 - p_{\text{surv}}(T, 0; h_0).$$

Proof. Step 1: resolvent equation. Taking the Laplace transform of the PIDE $\partial_t p_{\text{surv}} = \mathcal{L}^{(0)} p_{\text{surv}}$ with initial condition $p_{\text{surv}}(0, z; h_0) = 1$ and killing condition $p_{\text{surv}} = 0$ for $z \le -h_0$, exactly as in Section 5 with $g_k \equiv 1$, gives

$$(q - \mathcal{L}^{(0)})\widehat{U}_{\text{surv}}(q, z; h_0) = 1, \qquad z > -h_0.$$

Step 2: particular solution. The constant forcing $f \equiv 1$ is an eigenfunction of $\mathcal{L}^{(0)}$ with eigenvalue $\psi_Z^{(0)}(0) = 0$, so the particular solution is

$$\widehat{U}_{\text{surv,part}}(q, z) = \frac{1}{q - \psi_Z^{(0)}(0)} = \frac{1}{q}.$$

Step 3: homogeneous solution. By Lemma 5.2 applied with $k = 0$, the admissible homogeneous solution on $z > -h_0$ is

$$\widehat{U}_{\text{surv,hom}}(q, z; h_0) = C_1 e^{\gamma_4^{(0)}(q)(z+h_0)} + C_2 e^{\gamma_5^{(0)}(q)(z+h_0)} + C_3 e^{\gamma_6^{(0)}(q)(z+h_0)},$$

where $\gamma_4^{(0)}, \gamma_5^{(0)}, \gamma_6^{(0)}$ are the roots of $\psi_Z^{(0)}(\gamma) = q$ with $\text{Re}(\gamma) < 0$.

Step 4: boundary conditions. The killing condition imposes, at $z = -h_0^+$:

(i) $\widehat{U}_{\text{surv}}(q, -h_0^+; h_0) = 0$;

(ii) $K_{j,-}^{(0)}[\widehat{U}_{\text{surv}}](-h_0^+) = 0, \quad j = 1, 2,$

by the same argument as in Proposition 5.1: downward jumps from $-h_0^+$ land in the killed region.

Step 5: linear system. Substituting the full solution $\widehat{U}_{\text{surv}} = 1/q + \widehat{U}_{\text{surv,hom}}$ into each condition and applying (10) to the homogeneous terms yields

$$\mathbf{M}_{\text{surv}}(q)\, \mathbf{C}(q; h_0) = -\begin{pmatrix} 1/q \\ 1/q \\ 1/q \end{pmatrix},$$

where the right-hand side entries are $K_{j,-}^{(0)}[1/q](-h_0^+) = 1/q$ (since $K_{j,-}^{(0)}$ integrates to one against the exponential density). By Lemma 5.4 below, $\mathbf{M}_{\text{surv}}(q)$ is invertible, so the coefficients $\mathbf{C}$ are uniquely determined.

Step 6: inversion and liquidation probability. Evaluating at $Z_0 = 0$ and inverting via the Bromwich formula gives $p_{\text{surv}}(T, 0; h_0)$. The liquidation probability follows from the complement. □

Lemma 5.4 (Invertibility of $\mathbf{M}_{\text{surv}}(q)$)**.** Under Assumption 5.1 and condition (18), the survival matrix $\mathbf{M}_{\text{surv}}(q)$ of Proposition 5.2 is invertible.

Proof. The argument is identical to that of Lemma 5.3 with $k = 0$. The phase rates are $r_{1,-}^{(0)} = 1/\eta_{1,-}$ and $r_{2,-}^{(0)} = 1/\eta_{2,+}$, so condition (18) is equivalent to $\eta_{1,-} \neq \eta_{2,+}$. Each denominator factor is nonzero because $-r_{j,-}^{(0)}$ is a pole of $\psi_Z^{(0)}$, hence cannot equal any root $\gamma_m^{(0)}(q)$ of $\psi_Z^{(0)}(\cdot) = q$ for finite q. □

6 Numerical Results

We validate the Laplace–resolvent method against Monte Carlo (MC) simulation and verify the analytical no-shorting limit as $h_0 \to \infty$. Time-domain quantities are recovered via the Talbot deformed-contour algorithm (Talbot, 1979). No Bromwich shift needs to be chosen by hand.

The Kou model and contract parameters are given in Table 3. The two assets are assigned distinct downward-jump means ($\eta_{1,-} = 0.08$, $\eta_{2,-} = 0.12$), which ensures the phase rates $r_{1,-}^{(0)} \neq r_{2,-}^{(0)}$ and the barrier matrix $\mathbf{M}_{\text{bar}}^{(k)}$ is non-singular for all admissible k (here $k_{\max} = \lfloor 1/0.09 \rfloor - 1 = 10$).

For each $h_0 \in \{0.01, 0.10, 0.25, 0.50, 0.75, 1.00, 1.50, 2.00\}$ we report the survival probability p_{surv} and the following conditional statistics of $\Pi_T \mid \{\tau > T\}$, where $\mu_k := \mathbb{E}[\Pi_T^k \mid \tau > T]$:

$$\text{mean} := \mu_1, \quad \text{variance} := \mu_2 - \mu_1^2, \quad \text{skewness} := \frac{\mu_3 - 3\mu_1\mu_2 + 2\mu_1^3}{(\mu_2 - \mu_1^2)^{3/2}}, \quad \text{ex. kurt} := \frac{\mu_4 - 4\mu_1\mu_3 + 6\mu_1^2\mu_2 - 3\mu_1^4}{(\mu_2 - \mu_1^2)^2} - 3. \tag{19}$$

The MC benchmark uses 10^5 Euler–Poisson paths with 252 steps and seed 42; conditional statistics are computed over surviving paths only.

Group	Parameter	Asset 1	Asset 2
Kou	Expected price growth μ_i	0.10	0.08
	Diffusion volatility σ_i	0.30	0.25
	Jump intensity λ_i	2.0	2.0
	Upward-jump probability p_i	0.50	0.50
	Mean upward-jump size $\eta_{i,+}$	0.10	0.09
	Mean downward-jump size $\eta_{i,-}$	0.08	0.12
Correlation / contract	Brownian correlation ρ	0.50	
	Collateral factor b	0.80	
Portfolio	Initial prices $S_{1,0} = S_{2,0}$	1.00	
	Horizon T	1.00	

Table 3: Kou model and contract parameters. Derived quantities: $\chi_1 \approx 0.0185$, $\chi_2 \approx 0.0172$, $\mu_1^{\text{eff}} \approx 0.0180$, $\mu_2^{\text{eff}} \approx 0.0092$.

	p_{surv}		Mean		Variance	
h_0	Laplace	MC	Laplace	MC	Laplace	MC
0.01	0.0271	0.0550	3.4548	3.4362	3.5915	3.7417
0.10	0.2420	0.2631	2.4707	2.4436	1.8218	1.8338
0.25	0.5290	0.5444	1.7329	1.7200	0.9009	0.9052
0.50	0.8331	0.8412	1.2922	1.2862	0.4955	0.4996
0.75	0.9560	0.9586	1.1602	1.1590	0.3648	0.3671
1.00	0.9908	0.9917	1.1231	1.1235	0.2975	0.2990
1.50	0.9998	0.9998	1.1102	1.1113	0.2289	0.2299
2.00	1.0000	1.0000	1.1078	1.1090	0.1992	0.2001

Table 4: Survival probability, conditional mean and conditional variance: Laplace–resolvent vs. Monte Carlo (10^5 paths, 252 steps). The discrepancy in p_{surv} at $h_0 = 0.01$ reflects the scarcity of surviving paths ($\approx$2,700 out of 10^5).

Note on $h_0 = 0.01$. At $h_0 = 0.01$ the survival probability is only $\approx$2.7%, leaving roughly 2,700 surviving paths out of 10^5. MC estimates of higher-order statistics are therefore noisy at this point; the Laplace method is unaffected.

No-shorting limit. As $h_0 \to \infty$ the short holding $n_2 \to 0$ and the portfolio converges to a unit long position in asset 1 (Remark 4.1). All conditional statistics converge to those of $e^{X_{1,T}}$ under $\mathbb{P}$, which are available in closed form from the characteristic exponent (8):

$$\lim_{h_0 \to \infty} \mu_k = \mathbb{E}_{\mathbb{P}}\big[e^{kX_{1,T}}\big] = \exp\big(T\,\Psi(-ik, 0)\big).$$

Table 6 compares the Laplace output at $h_0 = 100$ against these targets. The relative errors are below 10^{-11}, confirming the internal consistency of the resolvent construction, the Talbot algorithm, and the drift normalisation $\mu_i^{\text{eff}} = \mu_i - \frac{1}{2}\sigma_i^2 - \lambda_i \chi_i$.

	Skewness		Excess kurtosis		CPU time (s)	
h_0	Laplace	MC	Laplace	MC	Laplace	MC
0.01	2.030	2.101	9.33	8.01	0.06	3.34
0.10	2.006	2.036	9.09	8.28	0.06	3.34
0.25	1.946	1.962	8.51	7.87	0.06	3.34
0.50	1.784	1.803	7.20	6.97	0.06	3.30
0.75	1.625	1.645	6.18	6.06	0.06	3.28
1.00	1.539	1.562	5.72	5.65	0.06	3.29
1.50	1.502	1.527	5.48	5.44	0.06	3.28
2.00	1.490	1.517	5.35	5.33	0.06	3.33

Table 5: Conditional skewness, excess kurtosis, and wall-clock CPU time: Laplace–resolvent vs. Monte Carlo (10^5 paths, 252 steps). Laplace time covers five Talbot inversions ($k = 0, 1, 2, 3, 4$) on a standard laptop. The Laplace method achieves a $\approx$55$\times$ speed-up.

Statistic	Analytical	Laplace ($h_0 = 100$)	|rel. error|
Mean	1.10517092	1.10517092	$< 10^{-11}$
Variance	0.16677912	0.16677912	$< 10^{-11}$
Skewness	1.44931133	1.44931133	$< 10^{-10}$
Excess kurtosis	5.03708463	5.03708463	$< 10^{-10}$

Table 6: No-shorting limit: Laplace at $h_0 = 100$ vs. analytical statistics of $e^{X_{1,T}}$ under $\mathbb{P}$. The analytical values use raw moments $\mathbb{E}[e^{kX_{1,T}}] = \exp(T\,\Psi(-ik, 0))$ converted via (19).

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A Generalised Beta of the Second Kind Approximation to the Payoff Distribution

The semi-analytic machinery of Section 5 delivers the first four killed moments $M_k(h_0)$, $k = 1, 2, 3, 4$, and the survival probability $p_{\text{surv}}(h_0)$. This appendix shows how to fit a four-parameter *Generalised Beta of the Second Kind* (GB2) distribution to these outputs.

Structure of the conditional law. On $\{\tau > T\}$ the payoff satisfies

$$Y := \Pi_T \mid \{\tau > T\} = \frac{n_2}{b}\, S_{2,0}\, e^{X_{2,T}}\, (e^{h_0+Z_T} - b).$$

Since $Z_T > -h_0$ on $\{\tau > T\}$, we have $e^{h_0+Z_T} > 1 > b$, so the factor $(e^{h_0+Z_T} - b) > 0$. However, because $e^{X_{2,T}}$ can be arbitrarily close to zero, $\inf Y = 0$, and the support is $Y \in (0, \infty)$ with no natural positive shift. The full payoff law is

$$\Pi_T \overset{d}{=} p_{\text{liq}}\, \delta_0 + p_{\text{surv}}\, Y, \tag{20}$$

where $p_{\text{liq}} = 1 - p_{\text{surv}}$.

Conditional survival moments. Since $\Pi_T^k = Y^k\, \mathbf{1}_{\{\tau>T\}}$,

$$\mu_k^+ := \mathbb{E}[Y^k] = \frac{M_k(h_0)}{p_{\text{surv}}(h_0)}, \qquad M_k(h_0) = \frac{e^{T\Psi(0,-ik)}}{(e^{h_0} - b)^k}\, m_k(T; h_0), \tag{21}$$

from Proposition 4.1, for $k = 1, 2, 3, 4$.

GB2 distribution. We model $Y \sim \text{GB2}(a, \beta, p, q)$ with density

$$f_Y(y) = \frac{a\, y^{ap-1}}{\beta^{ap}\, B(p,q)\, \big(1 + (y/\beta)^a\big)^{p+q}}, \qquad y > 0, \tag{22}$$

where $a > 0$ is a tail/shape parameter, $\beta > 0$ is a scale parameter, $p > 0$ and $q > 0$ are left- and right-shape parameters, and $B(p, q) = \Gamma(p)\Gamma(q)/\Gamma(p + q)$ is the beta function.

GB2 moments. For $k < aq$, the k-th raw moment is

$$\mathbb{E}[Y^k] = \beta^k\, \frac{B(p + k/a,\ q - k/a)}{B(p, q)} = \beta^k\, \frac{\Gamma(p + k/a)\, \Gamma(q - k/a)}{\Gamma(p)\, \Gamma(q)}. \tag{23}$$

The first four moments exist if and only if $q > 4/a$.

Calibration. We fit (a, β, p, q) by matching the four conditional moments $\mu_1^+, \ldots, \mu_4^+$.

Step 1: shape parameters from scale-free ratios. Define

$$R_k := \frac{\mu_k^+}{(\mu_1^+)^k} = \frac{B(p + k/a,\ q - k/a)/B(p, q)}{\big[B(p + 1/a,\ q - 1/a)/B(p, q)\big]^k}, \qquad k = 2, 3, 4. \tag{24}$$

The ratios R_k depend only on (a, p, q), not on β. Solve

$$R_k(a, p, q) = \hat{R}_k, \qquad k = 2, 3, 4, \tag{25}$$

where $\hat{R}_k = \mu_k^+/(\mu_1^+)^k$ are the targets, subject to $a > 0$, $p > 0$, $q > 4/a$. For numerical stability, evaluate (24) in log-space:

$$\log \mathbb{E}[Y^k] = k \log \beta + \log \Gamma\big(p + \tfrac{k}{a}\big) + \log \Gamma\big(q - \tfrac{k}{a}\big) - \log \Gamma(p) - \log \Gamma(q). \tag{26}$$

Step 2: scale from the first moment. Given (a, p, q) from Step 1, recover

$$\beta = \mu_1^+ \, \frac{B(p, q)}{B(p + 1/a,\ q - 1/a)}. \tag{27}$$

Full distribution and quantiles. The CDF of Π_T is

$$F_\Pi(y) = \begin{cases} 0 & y < 0, \\ p_{\text{liq}} & y = 0, \\ p_{\text{liq}} + p_{\text{surv}}\, F_Y(y) & y > 0, \end{cases}$$

where F_Y is the GB2 CDF. For a confidence level $u \in (0, 1)$, the Value-at-Risk is

$$\text{VaR}(u) = \begin{cases} 0 & u \le p_{\text{liq}}, \\ F_Y^{-1}\!\left(\dfrac{u - p_{\text{liq}}}{p_{\text{surv}}}\right) & u > p_{\text{liq}}, \end{cases}$$

where F_Y^{-1} is the GB2 quantile function.

Consistency check. After fitting, verify that

$$\beta^k \, \frac{\Gamma(p + k/a)\, \Gamma(q - k/a)}{\Gamma(p)\, \Gamma(q)} = \mu_k^+, \qquad k = 1, 2, 3, 4.$$

Rationale for GB2. The GB2 family nests several common distributions (Burr XII, log-logistic, Pareto, Weibull, log-normal in a limit) and has four free parameters, enabling the simultaneous matching of four conditional moments. It accommodates the power-law tails that arise in Kou jump-diffusion models, which the Generalized Gamma family — having thinner stretched-exponential tails — cannot capture. The condition $q > 4/a$ provides an explicit testable constraint on whether the fourth moment of the conditional payoff is finite.